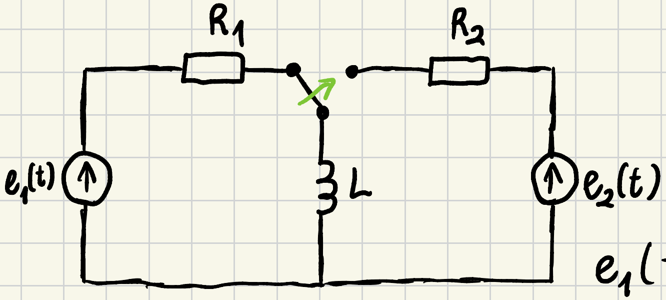
ĆWICZENIA 13

METODA OPERATOROWA

Zad. 1 FILIPOWICZ 6.12



$R_1 = 40\ \Omega$

$R_2 = 20\ \Omega$

$L = 0{,}4\ H$

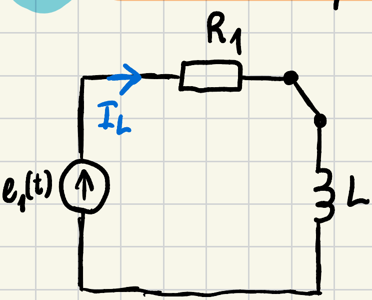
W „oryginale" jest 240

$e_1(t) = 200 \sin(100t + 90°)\ V$

a) $e_2(t) = 100\ V$

b) $e_2(t) = 140\sqrt{2} \sin(50t + 135°)\ V$

a) 1° Warunki początkowe



Najpierw musimy wyznaczyć prąd cewki $i_L(0^-)$ przed przełączeniem!

$\omega_1 = 100\ \frac{rad}{s}$ $\quad E_1 = \frac{200}{\sqrt{2}} e^{j90°} = 100\sqrt{2}\, e^{j90°}$

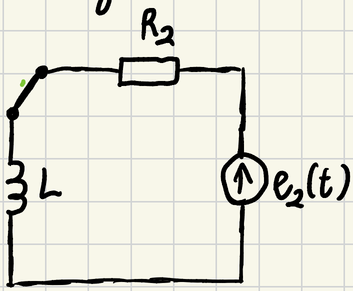
$Z_z = R_1 + Z_L = R_1 + j\omega_1 L = 40 + j40 = 40\sqrt{2}\, e^{j45°}$

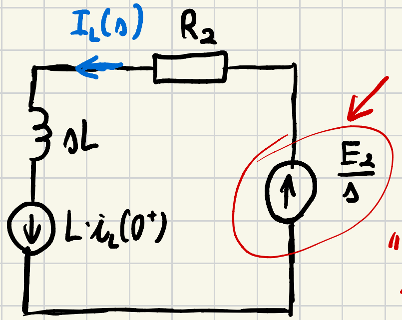
$I_L = \frac{E_1}{Z_z} = \frac{100\sqrt{2}\, e^{j90°}}{40\sqrt{2}\, e^{j45°}} = 2{,}5\, e^{j45°}$

$i_L(t) = 2{,}5\sqrt{2} \sin(100t + 45°)$

$i_L(0^-) = 2{,}5\ [A]$

2° Wyznaczanie $i_L(t)$





$e_2(t) \sim const.$ Dlatego możemy „przyśpieszyć" proces A1.

$E_2 = 100\ V$

$L \cdot i_L(0^+) = L \cdot i_L(0^-) = 0{,}4 \cdot 2{,}5 = 1\ [V]$

$i_L(0^-) = i_L(0^+) \rightarrow$ wynika z prawa komutacji cewki

$$I_L(s) = \frac{\frac{E_2}{s} + L i_L(0^+)}{R_2 + sL} = \frac{\frac{100}{s} + 1}{20 + 0{,}4s} = \frac{s + 100}{s(20 + 0{,}4s)} =$$

$$= \frac{s + 100}{0{,}4s(s+50)} = \frac{2{,}5s + 250}{s(s+50)} \begin{array}{l} \rightarrow s_1 = 0 \\ \rightarrow s_2 = -50 \end{array}$$

$$f(t) = \lim_{s \to s_1} \left[F(s) \cdot (s - s_1) e^{s_1 t} \right]$$

$$\frac{2{,}5 \cdot 0 + 250}{\cancel{s}_1(0 + 50)} \cdot (\cancel{s-0})^{1} \cdot e^{0t}$$

$$i_L(t) = \frac{2{,}5 \cdot 0 + 250}{\cancel{s}(0+50)} (\cancel{s-0}) \cdot e^{0t} + \frac{2{,}5 \cdot (-50) + 250}{-50\,\cancel{(s+50)}} \cancel{(s+50)} e^{-50t} =$$

$$= 5 - 2{,}5e^{-50t}$$

$$\boxed{i_L(t) = 5 - 2{,}5e^{-50t}}$$

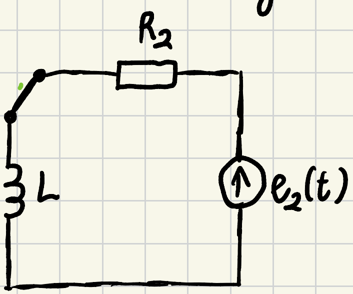
A1. „Przyśpieszamy", bo jeśli nie ma źródła sinusoidalnego po przełączeniu to nie ma konieczności liczenia stanu ustalonego

b) **Warunki początkowe takie same :** $i_L(0^-) = 2{,}5\ A$

$$e_2(t) = 140\sqrt{2}\sin(50t + 135°)\ [V]$$

Źródło zmienne (sinusoidalne), więc musimy zastosować superpozycję (czyli podejście „schematyczne")

2° Stan ustalony obwodu po przełączeniu



$$R_2 = 20\ \Omega \quad L = 0{,}4\ H$$

$$\omega_2 = 50\ \frac{rad}{s} \quad E_2 = 140e^{j135°}$$

$$Z_2 = R_2 + Z_L = 20 + j\overset{L}{0{,}4}\cdot\overset{\omega_2}{50} =$$

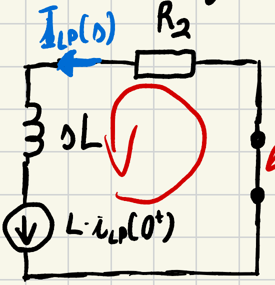
$$= 20 + j20 = 20\sqrt{2}e^{j45°}$$

$$I_L = \frac{E_2}{Z_2} = \frac{140e^{j135°}}{20\sqrt{2}e^{j45°}} = \frac{7}{\sqrt{2}}e^{j90°}$$

$$i_{Lu}(t) = 7\sin(50t + 90°)$$

$$\boxed{i_{Lu}(0^+) = 7\ A}$$

3° Stan przejściowy



Zwarcie, bo było tu źródło napięciowe E_2

$$i_L(t) = i_{Lu}(t) + i_{Lp}(t)$$

$$i_{Lp}(0^+) = i_L(0^-) - i_{Lu}(0^+)$$

$$i_{Lp}(0^+) = 2,5 - 7 = -4,5 \ [A]$$

$$L \cdot i_{Lp}(0^+) = 0,4 \cdot (-4,5) = -1,8 \ [A]$$

$$-4,5 \frac{1}{s+50}$$

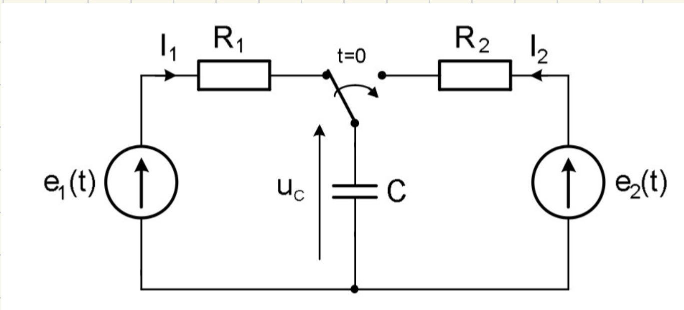
$$I_{LP}(s) = \frac{L \cdot i_{LP}(0^+)}{R_2 + sL} = \frac{-1,8}{20+0,4s} = \frac{-4,5}{s+50} \rightarrow s_1 = -50$$

$$i_{LP}(t) = \frac{-4,5}{\cancel{(s+50)}} \cancel{(s+50)} \cdot e^{-50t} = -4,5e^{-50t}$$

$$i_L(t) = i_{Lu}(t) + i_{LP}(t) = \boxed{7 \sin(50t + 90^\circ) - 4,5e^{-50t}}$$

$$\frac{s^2+4s+1}{s+1} = A + \frac{s+2}{s+1}$$

Zad. 2 FILIPOWICZ 6.11.



$e_1 = 200 \sin(100t + 90°)$ V

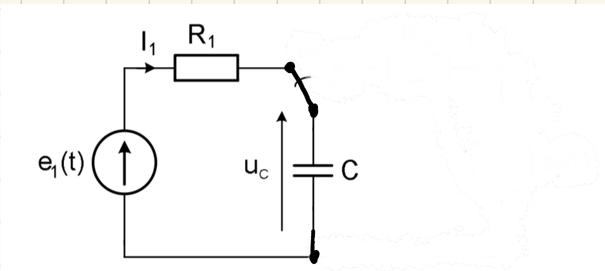
$R_1 = 50\ \Omega \quad R_2 = 10\ \Omega$

$C = 0{,}2\ \text{mF} = 2\cdot 10^{-4}$ F

a) $e_2(t) = 300$ V

b) $e_2(t) = 500\sqrt{2}\sin(500t + 135°)$ V

1° Warunki początkowe



$E_1 = \frac{200}{\sqrt{2}} e^{j90°} = 100\sqrt{2}\, e^{j90°}$

$\omega_1 = 100\ \frac{\text{rad}}{s}$

$X_C = \frac{1}{\omega_1 C} = \frac{1}{10^2 \cdot 2\cdot 10^{-4}} = 50\ \Omega$

$Z_C = -jX_C = -j50\ \Omega$

$Z_z = R_1 - jX_C = 50 - j50 = 50\sqrt{2}\, e^{-j45°}$

$I_1 = \frac{E_1}{Z_z} = \frac{100\sqrt{2}\, e^{j90°}}{50\sqrt{2}\, e^{-j45°}} = 2e^{j135°}$

$U_C = I_1 \cdot Z_C = 2e^{j135°} \cdot (-j50) = 100e^{j45°}$

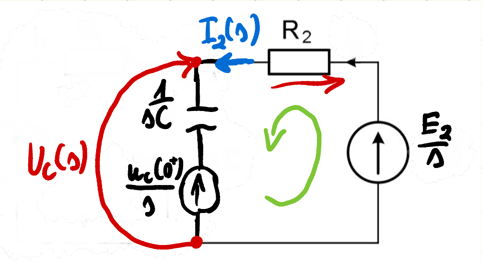
$u_C(t) = 100\sqrt{2}\sin(100t + 45°)$

$u_C(0^-) = 100 \cdot \sqrt{2} \cdot \frac{\sqrt{2}}{2} = 100$ V

$u_C(0^-) = 100$ V

2° Wyznaczanie $u_c(t)$

„Przyśpieszenie", bo $e_2(t) \sim const.$



$$E_2 = 300\,V$$

$$\frac{1}{sC} = \frac{1}{s \cdot 2 \cdot 10^{-4}} = \frac{5000}{s}$$

$$\frac{u_c(0^+)}{s} = \frac{u_c(0^-)}{s} = \frac{100}{s}$$

$$\frac{E_2}{s} - I_2(s)R_2 - I_2(s)\cdot\frac{1}{sC} - \frac{u_c(0^+)}{s} = 0$$

$$I_2(s)\left[R_2 + \frac{1}{sC}\right] = \frac{E_2}{s} - \frac{u_c(0^+)}{s}$$

$$I_2(s) = \frac{\frac{E_2}{s} - \frac{u_c(0^+)}{s}}{R_2 + \frac{1}{sC}} = \frac{\frac{300}{s} - \frac{100}{s}}{10 + \frac{5000}{s}} = \frac{\frac{200}{s}}{10 + \frac{5000}{s}} = \frac{200}{10s + 5000} = \frac{20}{s+500}$$

$$U_c(s) = I_2(s)\cdot\frac{1}{sC} + \frac{u_c(0^+)}{s} = \frac{20}{s+500}\cdot\frac{5000}{s} + \frac{100}{s} = 10^5\left[\frac{1}{s(s+500)}\right] + \frac{100}{s}$$

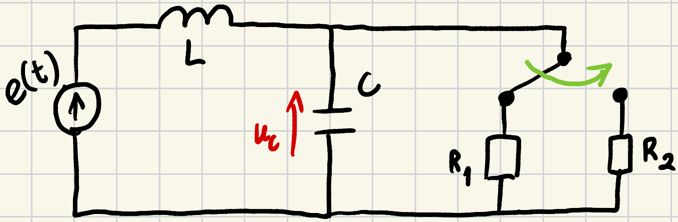
$$u_c(t) = 10^5\cdot\frac{1}{\cancel{s}(0+500)}\cancel{(s-0)}\cdot e^{0t} + 10^5\cdot\frac{1}{-500\cancel{(s+500)}}\cdot\cancel{(s+500)}\cdot e^{-500t} +$$

$$+ \frac{100}{\cancel{s}}\cdot\cancel{(s-0)}\, e^{0t}$$

$$u_c(t) = \frac{10^5}{500} + \frac{10^5}{-500}e^{-500t} + 100 = 200 - 200e^{-500t} + 100$$

$$\boxed{u_c(t) = 300 - 200e^{-500t}}$$

Zad. 3 JAKIŚ ??? ZBIÓR



$$e(t) = 10\ V \Rightarrow \omega = 0$$

$$R_1 = 5\ \Omega \quad R_2 = 50\ \Omega$$

$$L = 1\ H \quad C = 1000\ \mu F = 10^{3} \cdot 10^{-6} = 10^{-6}\ F$$

1° Warunki początkowe